| Task | Tuned Lens | | Logit Lens | | Baseline | Accuracy |
|---|---|---|---|---|---|---|
| | iForest | LOF | iForest | LOF | SRM | Normal → Injected |
| ARC-Easy | 0.59 (0.54, 0.62) | 0.73 (0.71, 0.76) | 0.53 (0.50, 0.57) | 0.59 (0.56, 0.62) | 0.73 (0.70, 0.75) | 72.8% → 31.7% |
| ARC-Challenge | 0.71 (0.65, 0.77) | 0.81 (0.77, 0.84) | 0.73 (0.67, 0.79) | 0.80 (0.77, 0.83) | 0.57 (0.53, 0.61) | 43.5% → 24.7% |
| BoolQ | 0.99 (0.98, 0.99) | 1.00 (1.00, 1.00) | 0.89 (0.87, 0.91) | 0.61 (0.57, 0.66) | 1.00 (1.00, 1.00) | 67.1% → 0.0% |
| MC TACO | 0.74 (0.71, 0.77) | 0.68 (0.66, 0.70) | 0.68 (0.66, 0.69) | 0.55 (0.53, 0.59) | 1.00 (1.00, 1.00) | 0.40 → 0.06 F1 |
| MNLI | 0.98 (0.98, 0.99) | 1.00 (1.00, 1.00) | 0.95 (0.94, 0.96) | 1.00 (1.00, 1.00) | 1.00 (1.00, 1.00) | 54.3% → 0.0% |
| QNLI | 0.99 (0.99, 1.00) | 1.00 (1.00, 1.00) | 0.93 (0.92, 0.95) | 0.68 (0.63, 0.71) | 1.00 (1.00, 1.00) | 54.3% → 0.0% |
| QQP | 1.00 (0.99, 1.00) | 1.00 (1.00, 1.00) | 0.90 (0.89, 0.90) | 0.79 (0.76, 0.81) | 1.00 (1.00, 1.00) | 60.7% → 6.5% |
| SciQ | 0.62 (0.57, 0.69) | 0.64 (0.59, 0.70) | 0.75 (0.71, 0.79) | 0.70 (0.65, 0.74) | 0.75 (0.72, 0.78) | 95.5% → 62.6% |
| SST-2 | 1.00 (0.98, 1.00) | 1.00 (1.00, 1.00) | 0.78 (0.72, 0.83) | 0.61 (0.56, 0.65) | 1.00 (1.00, 1.00) | 82.9% → 49.1% |

*Table 1.* Test set AUROCs and 95% bootstrap CIs for distinguishing normal prompts from prompt injections on Pythia 12B. Figures are pooled over 10 random train-test splits. Attack detection performance is nearly perfect on tasks where the attack succeeds at driving accuracy well below the random baseline, and is still much better than chance even when the attack is only partially successful.

(Breunig et al., 2000), both implemented in scikit-learn (Pedregosa et al., 2011) with default hyperparameters.

**Baseline.** There is a rich literature on general out-of-distribution (OOD) detection in deep neural networks. One simple technique is to fit a multivariate Gaussian to the model's final layer hidden states on the training set, and flag inputs as OOD if a new hidden state is unusually far from the training distribution as measured by the Mahalanobis distance (Lee et al., 2018; Mahalanobis, 1936).

Recently, Bai et al. (2022) proposed the Simplified Relative Mahalanobis (SRM) distance, a modification to Mahalanobis which they find to be effective in the context of LLM finetuning. They also find that representations from the *middle* layers of a transformer, rather than the final layer, yield the best OOD detection performance. We use the SRM at the middle layer as a baseline in our experiments.

**Experimental setup.** We fit each anomaly detection model exclusively on prediction trajectories from *normal* prompts without prompt injections, and evaluate them on a held out test set containing both normal and prompt-injected trajectories. This ensures that our models cannot overfit to the prompt injection distribution. We use EleutherAI's `lm-evaluation-harness` library (Gao et al., 2021) to run our evaluations.

**Results.** Our results are summarized in Table 1. Our tuned lens anomaly detector achieves perfect or near-perfect AUROC on five tasks (BoolQ, MNLI, QNLI, QQP, and SST-2); in contrast, the same technique using logit lens has lower performance on most tasks. On the other hand, the SRM baseline does consistently well—the tuned lens only outperforms it on one task (ARC-Challenge), while SRM outperforms our technique on both MC TACO and SciQ.

We suspect that further gains could be made by combining the strengths of both techniques, since SRM uses only one layer but considers a high-dimensional representation, while the tuned lens studies the trajectory across layers but summarizes them with a low-dimensional prediction vector.

### 5.4. Measuring Example Difficulty

Early exiting strategies like CALM (Schuster et al., 2022) and DeeBERT (Xin et al., 2020) are based on the observation that "easy" examples require less computation to classify than "difficult" examples. If an example is easy, the model should quickly converge to the right answer in early layers, making it possible to skip the later layers without a significant drop in prediction quality. Conversely, the number of layers needed to converge on an answer can be used to measure the difficulty of an example.

We propose to use the tuned lens to estimate example difficulty in *pretrained* transformers, without the need to finetune the model for early exiting. Following Baldock et al. (2021)'s work on computer vision models, we define the *prediction depth* of a prompt $\boldsymbol{x}$ to be the number of layers after which a model's top-1 prediction for $\boldsymbol{x}$ stops changing.

To validate the prediction depth, we measure its correlation with an established difficulty metric: the *iteration learned*. The iteration learned is defined as the earliest training step $\tau$ where the model's top-1 prediction for a datapoint $\boldsymbol{x}$ is fixed (Toneva et al., 2018). Intuitively, we might expect that examples which take a long time to learn during training would tend to require many layers of computation to classify at inference time. Baldock et al. (2021) indeed show such a correlation, using k-NN classifiers to elicit early predictions from the intermediate feature maps of image classifiers.

**Experimental setup.** For this experiment we focus on Pythia 12B (deduped), for which 143 uniformly spaced checkpoints are available on Huggingface Hub. We evaluate the model's zero-shot performance on twelve multiple-choice tasks, listed in Table 2. For each checkpoint, we store the top 1 prediction on every individual example, allowing us to compute the iteration learned. We then use the tuned lens on the final checkpoint, eliciting the top 1 prediction at each layer of the network and computing the

| Task | Tuned lens $\rho$ | Logit lens $\rho$ | Final acc |
|---|---|---|---|
| ARC-Easy | **0.577** | 0.500 | 69.7% |
| ARC-Challenge | **0.547** | 0.485 | 32.4% |
| LogiQA | **0.498** | 0.277 | 21.4% |
| MNLI | 0.395 | **0.435** | 40.4% |
| PiQA | **0.660** | 0.620 | 76.1% |
| QNLI | **0.409** | −0.099 | 53.0% |
| QQP | **0.585** | −0.340 | 0.381 (F1) |
| RTE | 0.156 | **0.347** | 60.0% |
| SciQ | **0.530** | 0.505 | 91.9% |
| SST-2 | **0.555** | 0.292 | 64.7% |
| WinoGrande | 0.517 | **0.537** | 63.9% |

*Table 2.* Correlation between two measures of example difficulty, the iteration learned and prediction depth, across tasks. Prediction depth is measured using the tuned lens in the first column and the logit lens in the second.

prediction depth for every example. As a baseline, we also compute prediction depths using the logit lens. Finally, for each task, we compute the Spearman rank correlation between the iteration learned and the prediction depth across all examples.

**Results.** We present results in Table 2. We find a significant positive correlation between the iteration learned and the tuned lens prediction depth on all tasks we investigated. Additionally, the tuned lens prediction correlates better with iteration learned than its logit lens counterpart in 8 out of 11 tasks, sometimes dramatically so.

## 6. Discussion

In this paper, we introduced a new tool for transformer interpretability research, the *tuned lens*, which yields new qualitative as well as quantitative insights into the functioning of large language models. It is a drop-in replacement for the logit lens that makes it possible to elicit interpretable prediction trajectories from essentially any pretrained language model in use today. We gave several initial applications of the tuned lens, including detecting prompt injection attacks.

Finally, we introduced *causal basis extraction*, which identifies influential features in neural networks. We hope this technique will be generally useful for interpretability research in machine learning.

**Limitations and future work.** One limitation of our method is that it involves training a translator layer for each layer of the network, while the logit lens can be used on any pretrained model out-of-the-box. This training process, however, is quite fast: our code can train a full set of probes in under an hour on a single 8×A40 node, and further speedups are likely possible. We have also released tuned lens checkpoints for the most commonly used pre-

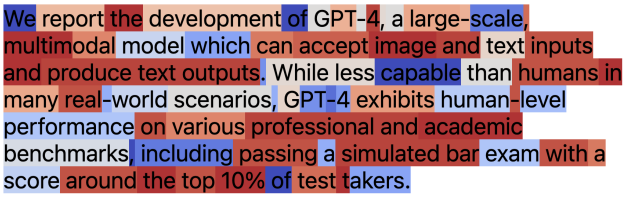


*Figure 12.* Token-level prediction depths for Pythia 12B computed on the abstract of OpenAI (2023). Warm colors have high prediction depth, while cool colors indicate low depth.

trained models as part of our `tuned-lens` library, which should eliminate this problem for most applications.

Causal basis extraction, as presented in this work, is computationally intensive, since it sequentially optimizes $d_{model}$ causal basis vectors for each layer of the network. Future work could explore ways to make the algorithm more scalable. One possibility would be to optimize a whole $k$-dimensional subspace, instead of an individual direction, at each iteration.

Due to space and time limitations, we focused on language models in this work, but we think it's likely that our approach is also applicable to other modalities.

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